

An Exploratory Analysis of Food Insecurity and Associated Factors in Metro Atlanta

Sofia Charlot & Tiyamika Williams

Undergraduate
Research Programs
Pathways Center

Abstract/Intro/Motivation

Food insecurity affects many Metro Atlanta neighborhoods, especially where structural barriers limit access to stable housing, transportation, and food assistance.

Using CDC PLACES data, we examined how food insecurity varies across census tracts, and which sociodemographic factors are most strongly associated with it.

Prior research highlights SNAP usage, housing instability, and transportation barriers as key contributors. Understanding these local patterns helps us think critically about how structural determinants shape food access and community health.

Our central research questions were:

- What do food insecurity levels look like across Metro Atlanta census tracts?
- How are housing insecurity, transportation barriers, and SNAP usage statistically associated with food insecurity?
- To what extent can these factors describe or predict food insecurity?

Methods/Approach

We analyzed census-tract indicators from CDC PLACES for Metro Atlanta. After loading and cleaning the data, we reviewed histograms, boxplots, and summary statistics for food insecurity, SNAP usage, housing insecurity, and transportation barriers.

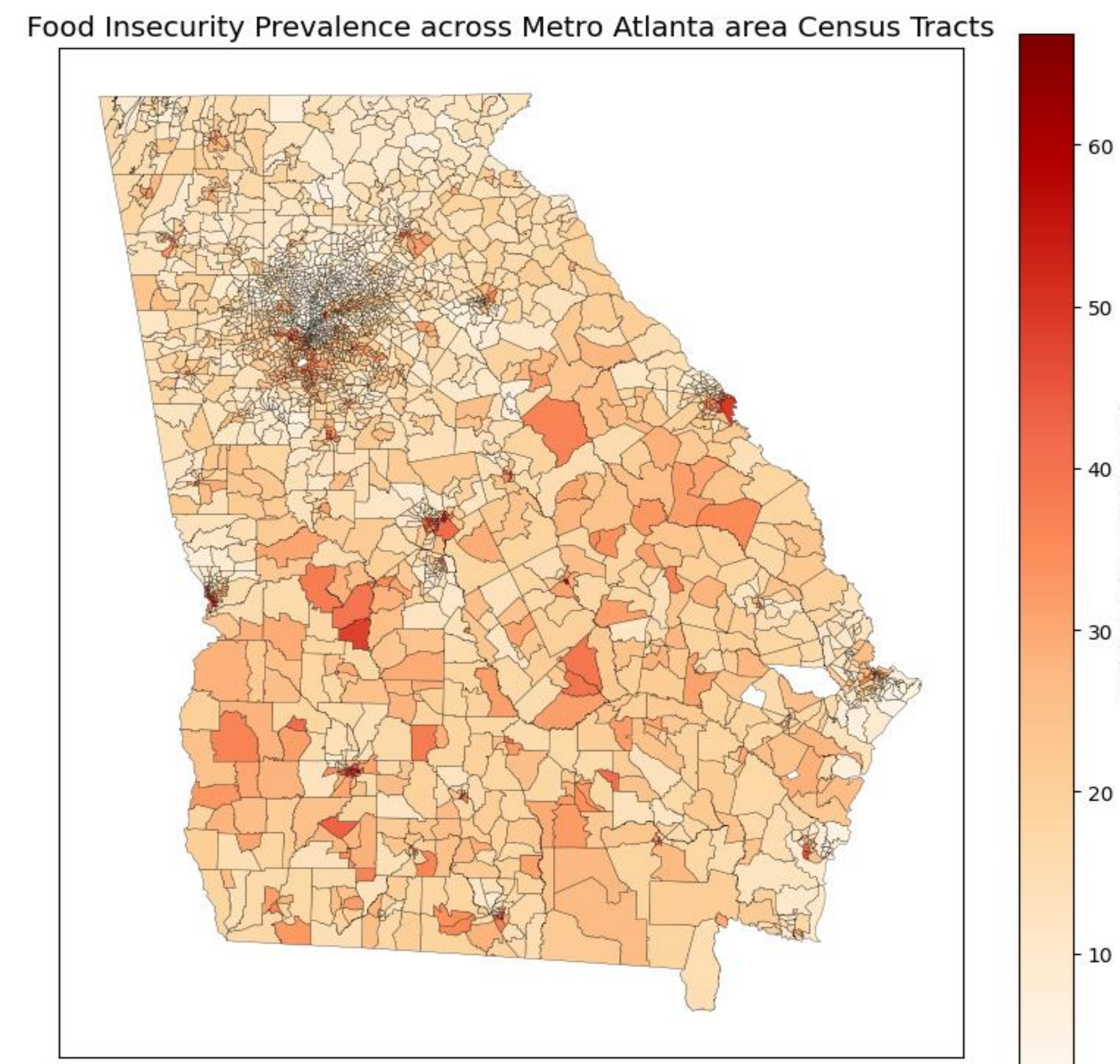
We used Pearson correlations to assess associations and ran simple linear regressions with food insecurity as the outcome and each sociodemographic variable as a predictor.

This approach allowed us to evaluate both descriptive patterns and predictive relationships.

Results

As seen in Figure 1, food insecurity is not evenly distributed across Metro Atlanta. Average food insecurity across all census-tracts is 18%. Certain tracts experience substantially higher (67% maximum) or lower food insecurity prevalence (3% minimum). This indicates disparities in food insecurity.

Figure 1. Food Insecurity Prevalence across the Metro Atlanta Area



Food insecurity was extremely strongly and positively correlated with SNAP usage ($r = 0.99$, $p < 0.001$), housing insecurity ($r = 0.98$, $p < 0.001$), and transportation barriers ($r = 0.99$, $p < 0.001$) at the census-tract level.

Figure 2 highlights strong positive relationships between food insecurity and structural factors such as SNAP usage, housing insecurity, and transportation barriers.

Conclusions/Future Directions

Taken together, this analysis underscores the need for public health interventions that consider place-based inequities. The geographic and structural patterns we identified point to opportunities for community partners, policy makers, and health systems to collaborate around solutions that improve stability, reduce barriers, and ultimately strengthen food access across Metro Atlanta.

Future work should analyze combined predictors, evaluate spatial clusters, and integrate community insights to guide policy and program development.

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Figure 2. Associations between Food Insecurity and Sociodemographic Factors

